# VOICE & AUDITORY INTERFACE ARCHITECTURE

Hypothetical research integration

**Consent-based research & simulation**

Concept architecture / Proposed links

External supply / battery

Controller power

Separate modality

Research EXG

Research host power

## NeuroSkill

Acquisition → Features → Embeddings → Annotations

Model estimates

## OPTIONAL CLOUD

API gateway → Authenticated transport → LLM endpoint → Optional TTS API → Model storage → Job queue

Ethernet / Wi-Fi

Cellular backhaul

Optional

WebSocket

## PROPOSED IMPLANT INTERFACES

### Acoustic sensing

Middle-ear motion

UmboMic → Charge amplifier → Acoustic digitizer → Device telemetry

### Neural sensing

Experimental

Recording array → Neural amplifier → Neural digitizer → Device telemetry

### DEVICE-SPECIFIC OUTPUT

Perceptual validation

L receiver / stimulator

R receiver / stimulator

L cochlear electrodes

R cochlear electrodes

Device power / charging

To be specified

## EXTERNAL CONTROLLER

Deployment alternatives

Controller power bus

### ACQUISITION & INTERPRETATION

ACOUSTIC

Device receiver → Time alignment → Signal quality → Audio features → Speech activity → Speech recognition → Recognized speech

Proposed adapter

NEURAL RESEARCH

Participant-specific

Experimental

Device receiver → Time alignment → Signal quality → Artifact rejection → Neural features → Task-trained decoder → Confidence gate

Streaming validation

Reject / no output

Candidate neural output

Reviewed settings

EXG context

Model estimates

### APPLICATIONS & ORCHESTRATION

Proposed adapter

NeuroLoop / pi

Planner adapter

Flowise

Event rules

Scenario library

Local / cloud LLM

State transitions

Task definition → Session state → Dialogue planner → Turn scheduler → Speech queue

Speech practice

Dialogue

Playback

One active turn

Synthetic-character profiles

Anonymous

Voice inventory

Scene size

History window

Configurable settings

Runtime context

MEMORY & RECORDS

Recorded events / Retrieval

Session history

Retrieval index

Scenario state

Voice registry

Model versions

Provenance

Retention

Audit log

### OUTPUT & PERCEPTION

Ear canal ← Speaker / headphones ← Acoustic playback

Auditory perception

Interface validation

Stimulation encoder ← Device speech processor ← CI audio interface

L

R

Compatible telemetry

Physical routes / Perceptual assessment

### VOICE CREATION & RENDERING

Runtime rendering

Offline preparation

Authorized recordings → Audio quality → Voice enrollment → Voice profile

Voice-owner consent

Model reference

Prosody

Pacing

Pronunciation

Voice ID

Optional mix inputs

Application text

Script text

Speaker-conditioned TTS

Selected music

Level

Ambient audio

Level

Live microphone

Voice profile

Voice conversion

Optional

Source selector

Audio chunks

Playback buffer

Turn arbiter

Approved clips

Clip playback

Experimental streaming

Speech renderer

Level limiter

Audio mix

Mute

Output monitor

Reviewed settings

Slower feedback

### CALIBRATION & VALIDATION

Device status → Link quality → Health monitor

Physical measurements

Measurement interface → Recorded response → Response analysis → Calibration review

Response sensing

Timing markers

User feedback → Perceptual assessment → Approved settings

Reported percepts

Output interrupt

Participant control

Start / stop

Session limits

Fault mute

### OFFLINE DEVELOPMENT

Labeled trials → Training → Held-out evaluation → Model registry → Approved update ← Rollback

Training datasets

Latency

Signal quality

Decoder error

Playback status

FUNCTION / DESIGN STATUS

Data

Audio

Control

Power

Proposed

Slow feedback

Schematic • Not to scale